PROTOTYPES FOR HUMANITY 2026 — APPLICATION

Closing the gap between education and employment in Africa.

SkillsSync is an AI platform that maps the skills inside students' coursework against real labor-market demand — turning four years of academic work into an evidence-based, employer-ready professional profile.

Project
SkillsSync

Category
Education & Employability · AI for Africa

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SDG Alignment
SDG 4 · SDG 8 · SDG 9 · SDG 17

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> "*The mismatch between education and skills and job requirements remains a key development challenge for African countries.*"
>
> — Morsy & Mukasa (2020), *African Development Review*, p. 1

01 · The Problem

> Across Sub-Saharan Africa, students invest four to six years and a significant share of household income in university degrees — only to graduate without the skills employers actually demand. By the time they discover the mismatch, the time to fix it is gone.

The numbers tell a brutal story.

28.9%
OF AFRICAN YOUTH ARE **UNDERSKILLED** FOR THE JOBS THEY HOLD

17.5%
ARE **OVERSKILLED** — CREDENTIALS DON'T MATCH WORK

3 in 4
AFRICAN YOUTH LACK SECURE, STABLE EMPLOYMENT (ILO 2024)

+76M
NEW SSA YOUTH IN THE LABOUR FORCE BY 2050 (ILO)

Why existing solutions don't work.

University career offices are reactive, not predictive

They engage students in their final semester — long after curriculum decisions have been made. The roadmap that matters is the one a student walks for four years, not the one they're handed at graduation.

CVs and transcripts are blind to demonstrated skill

A "B+ in Database Systems" tells an employer nothing about whether the student built a working query optimizer or copied a tutorial. The signal is buried in the coursework itself — never extracted, never indexed.

Generic LinkedIn-style platforms ignore Africa's reality

They assume English-language, digitally-typed work and stable internet. They cannot read the handwritten lecture notes, the photographed assignments, or the local-language project reports that make up most African student work.

The mismatch persists — it does not self-correct

Morsy & Mukasa's panel data shows skill mismatches in African labor markets are *persistent over time*. The problem cannot be left to "the market" — structural visibility tools are required.

> *The cost is not abstract: graduates with mismatched skills earn significantly less and report lower job satisfaction throughout their careers* *(Njifen, 2024, Economia Politica).*

02 · The Solution

> SkillsSync turns the work students are *already doing* — assignments, projects, lab reports, even handwritten lecture notes — into a structured, machine-readable record of demonstrated competencies, mapped against real labor-market demand. We give students, universities, and employers a shared, evidence-based view of what a degree actually delivers.

The core insight

Universities already collect millions of pages of evidence about what their students can do. That evidence sits inert in PDFs, photos, and notebooks. **SkillsSync extracts the signal, aligns it to a standardized skill taxonomy, and continuously closes the gap to a chosen career path.**

Evidence-based
Skills are extracted from real student artifacts — not self-reported. The output is auditable, citation-backed, and verifiable.

Africa-native
Designed for handwritten input, photographed work, and local-language documents — the formats African students actually produce.

Continuous, not one-shot
Each new document updates the student's competency graph and shrinks (or widens) the gap to their target role in real time.

What students get

- **A living skill profile** — every demonstrated competency, sourced to the exact document that proves it.
- **A personalized career roadmap** — the shortest path of skills, courses, and resources between where they are and the job they want.
- **A confidence score per skill** — so employers and the student can distinguish "named once in a slide" from "demonstrated across three projects."
- **A verifiable, on-chain credential** for each analyzed document — tamper-evident proof of work.

What universities get

- A cohort-level view of which skills the curriculum is actually producing — and which it claims to but doesn't.
- An early-warning system for at-risk graduating cohorts, course by course.
- Evidence to engage employers and accreditation bodies with concrete data.

03 · How It Works

> A student uploads a document. Five seconds later, every skill it demonstrates has been extracted, classified, mapped to the European ESCO taxonomy, and added to their cumulative competency graph. The roadmap to their chosen career path is recomputed automatically.

1 Upload
Photo, PDF, handwritten note, typed assignment — any format, any device, any data plan.

2 OCR & Layout
Transformer-based handwritten text recognition (TrOCR-class) parses pages into structured text + figures.

3 Skill Extraction
LLM identifies hard & soft skill spans, weighted by evidence strength and document type.

4 Taxonomy Mapping
Skills aligned to ESCO's 13,890-entry competency framework for cross-border comparability.

5 Roadmap
Shortest-path traversal across a skill graph computes the missing competencies for the student's target role.

End-to-end architecture

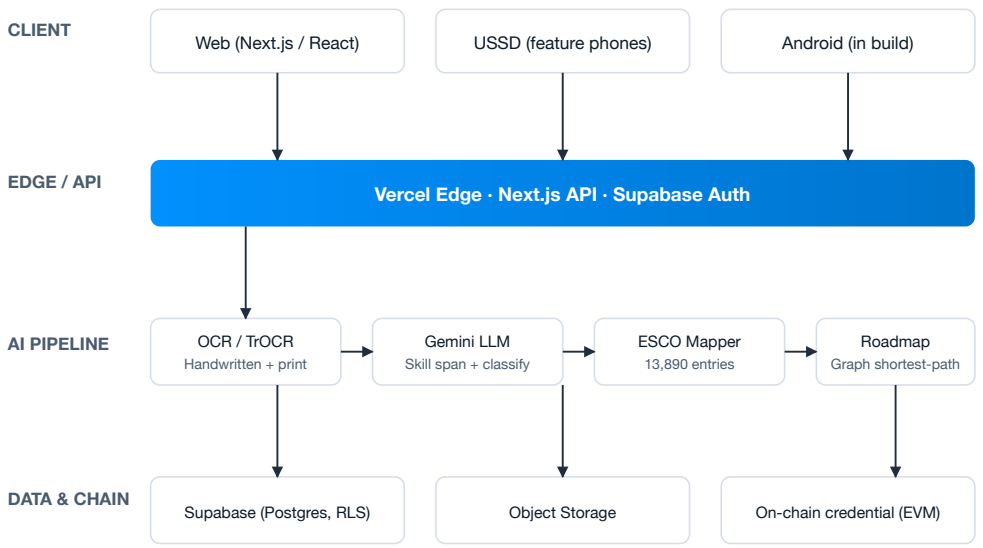


Figure 1 · SkillsSync end-to-end architecture across client, edge, AI pipeline and data layers.

03 · How It Works — inputs & outputs

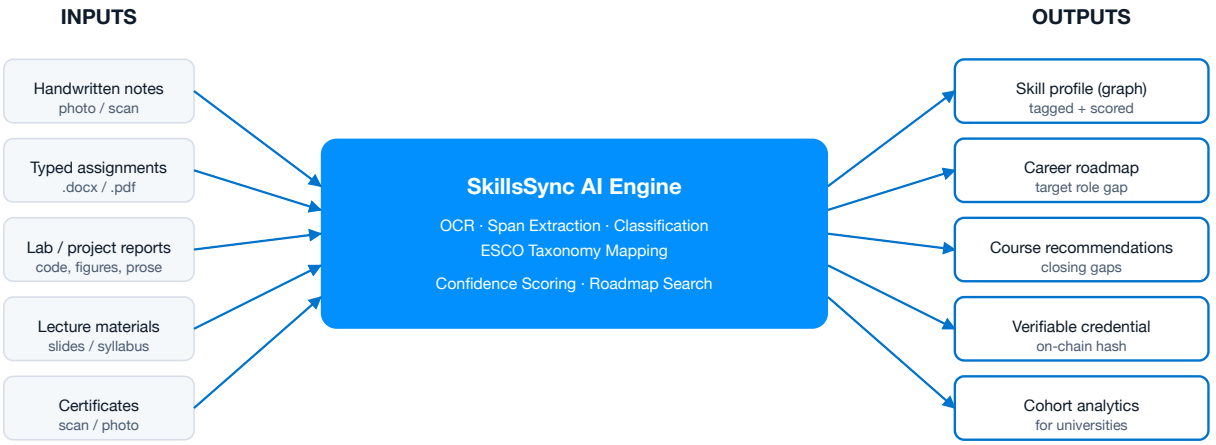


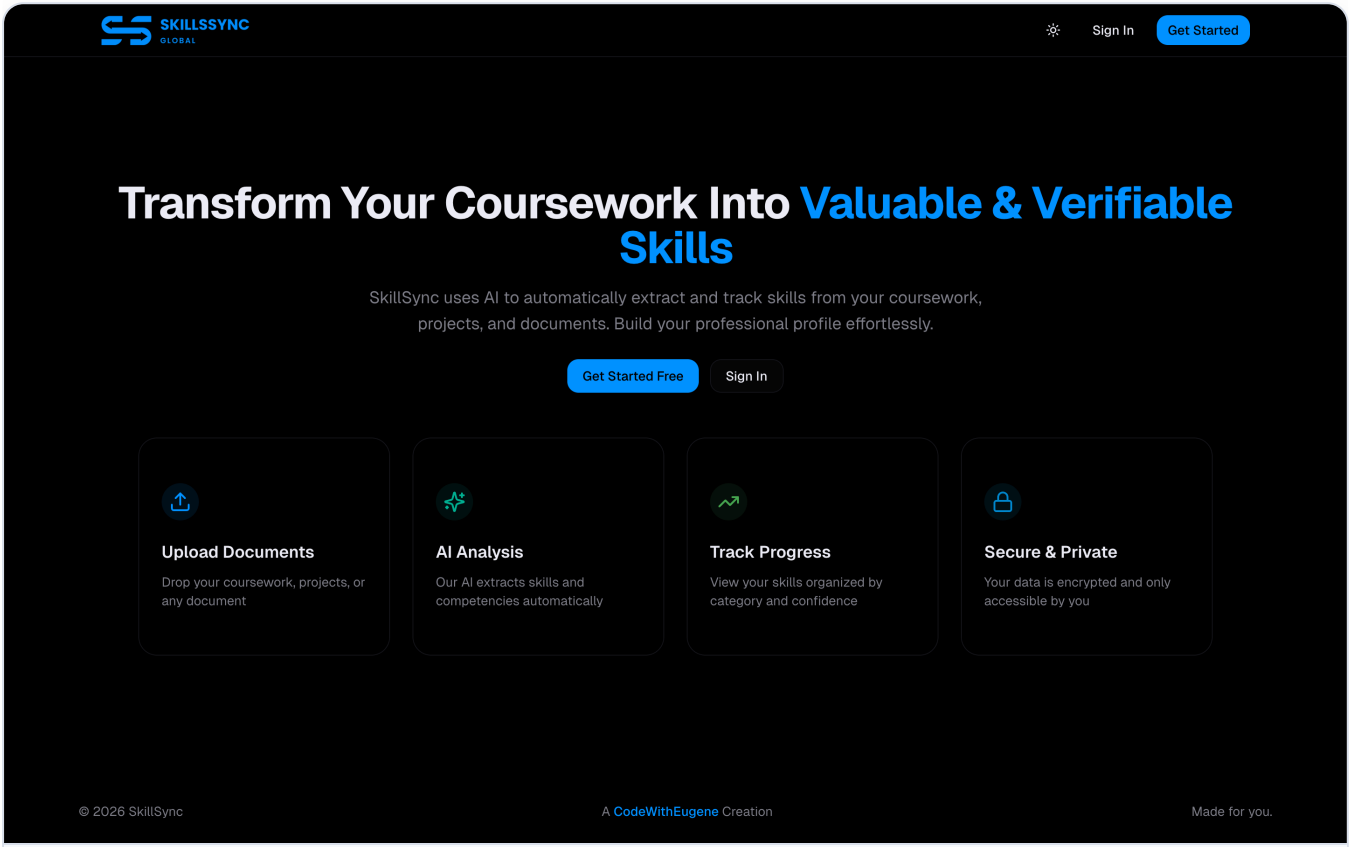
Figure 2 · Inputs the platform accepts and the structured artifacts it produces.

A walked example

A second-year computer science student at JKUAT uploads three things over a semester: a database design assignment (typed PDF), her hand-drawn ER diagrams (phone photo), and her end-of-semester project — a Python script with a brief written report. SkillsSync extracts **SQL design (high confidence)**, **data modeling (high)**, **Python programming (medium)**, **technical writing (medium)**, and **requirement analysis (low)**. Her chosen career — *data engineer* — requires nine ESCO competencies; she has six. The roadmap surfaces the missing three (Apache Airflow, dimensional modeling, cloud data warehouses) and recommends three targeted courses to close them before final year.

04 · The Product

> SkillsSync is live at www.skillssync.xyz. Below is the production landing page; subsequent pages illustrate the core dashboard surfaces students see after signing in.



Live production landing page · www.skillssync.xyz

Five product surfaces

1. Document upload & analysis

Drag-and-drop or photo-capture upload. Status moves through PROCESSING → COMPLETED in seconds. Each document gets a per-skill evidence breakdown linked back to the source page.

2. Skill profile

Skills clustered by category (technical, analytical, communication, leadership, domain), each with a confidence score and a list of source documents that demonstrate it.

3. Career roadmap

4. Insights & comparison

Choose a target role; the platform computes the shortest path of missing competencies, the order in which to learn them, and concrete resources to acquire each.

Student-level: how a profile compares to peers in the same career track. Institutional-level: cohort coverage of program learning outcomes — which the curriculum claims to deliver vs. what it actually demonstrably produces.

5. Goals & progress

Set sub-goals (master Pandas, build a portfolio site, ship an open-source PR). Document uploads and external course completions automatically update progress.

6. Verifiable credential

Each completed analysis produces a content-addressable hash recorded on-chain — students share a tamper-evident proof of work with employers, not just a self-reported skill list.

04 · The Product — student dashboard

Each authenticated student lands on a personalised dashboard surfacing four anchor metrics (documents, skills, completed analyses, in-flight analyses), their current career-readiness score, a per-document skill-growth timeline, and a live skill-gap list. Below is a screenshot from the live production app.

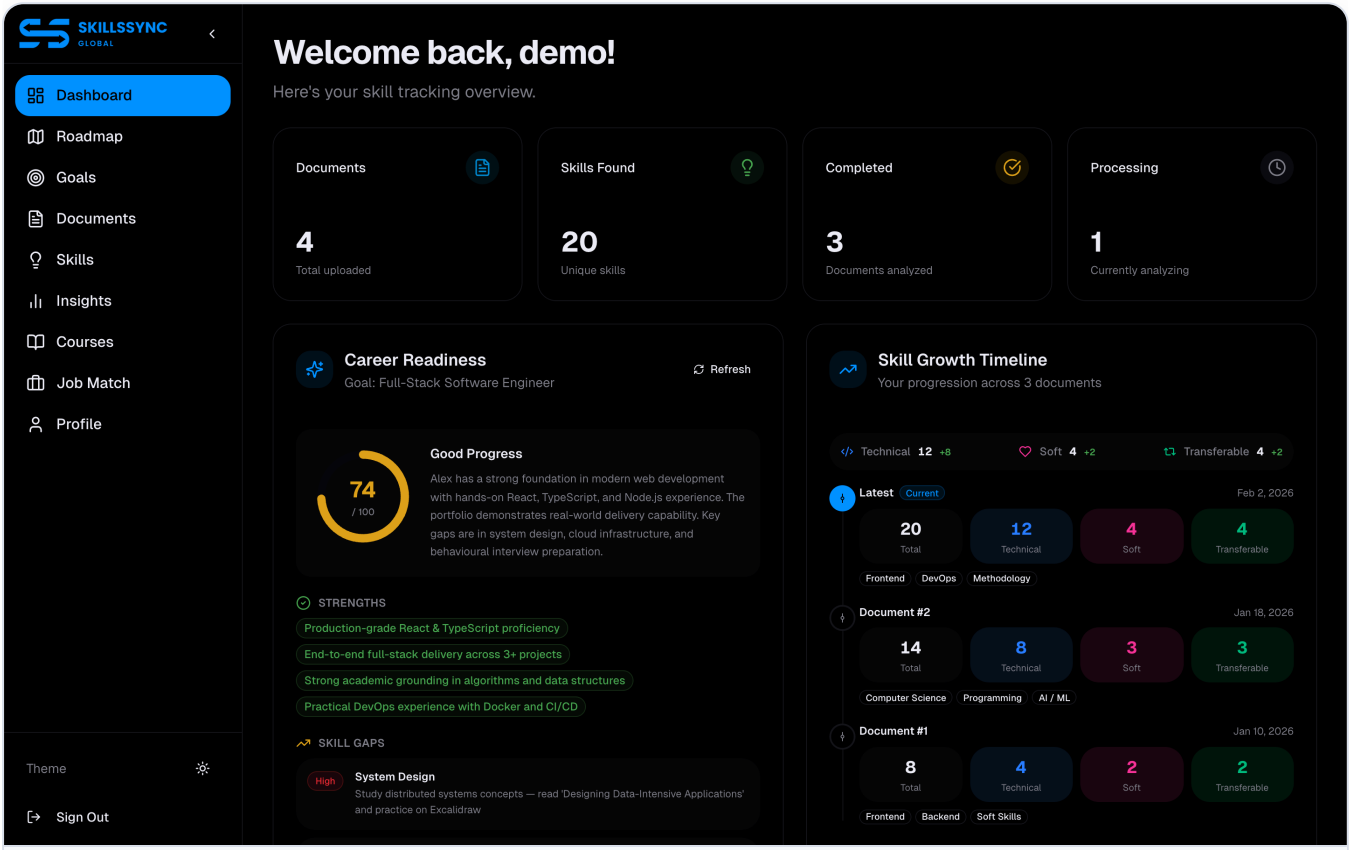


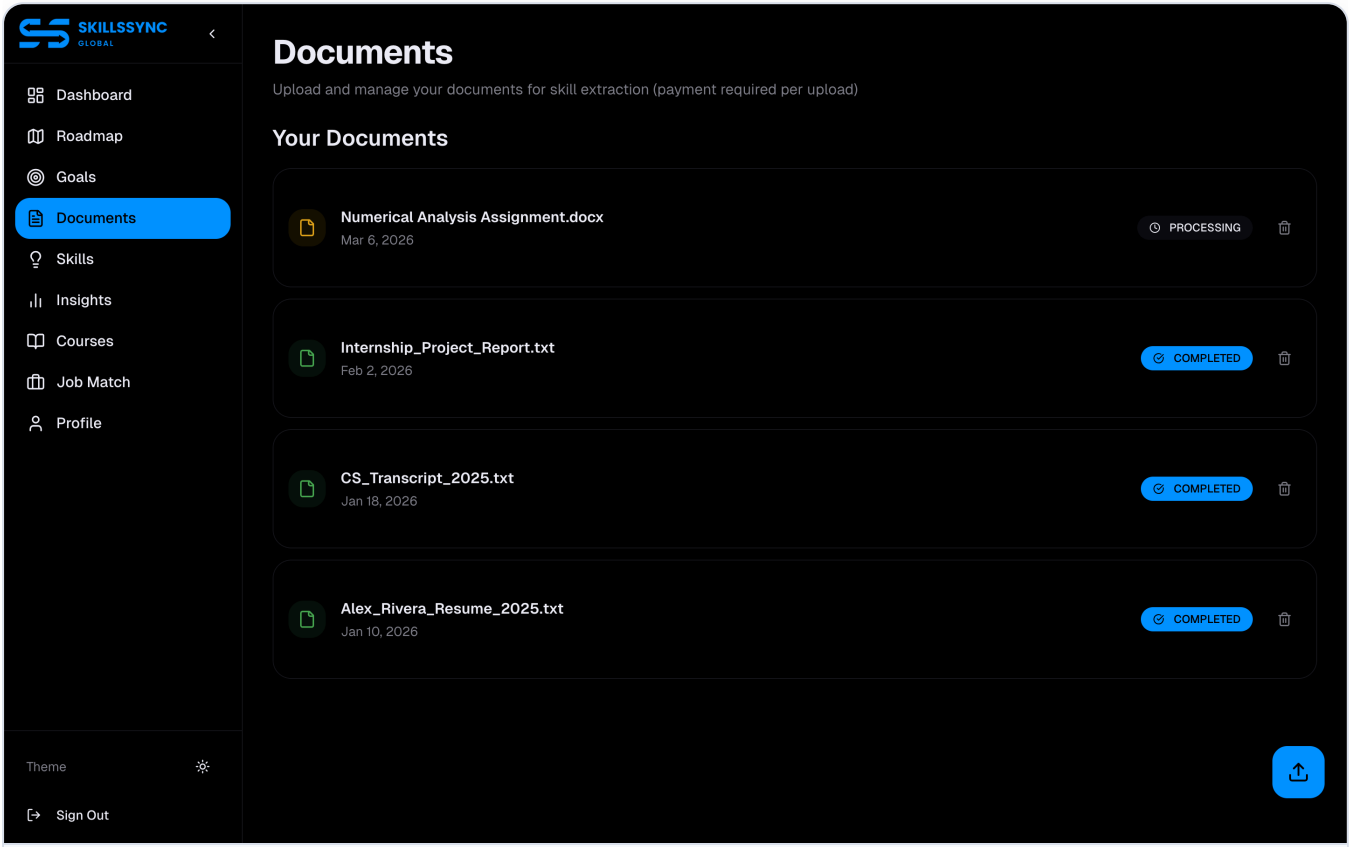
Figure 3 · Live student dashboard at app.skillssync.xyz — career-readiness, skill timeline, and recent documents.

What the dashboard tells the student in five seconds

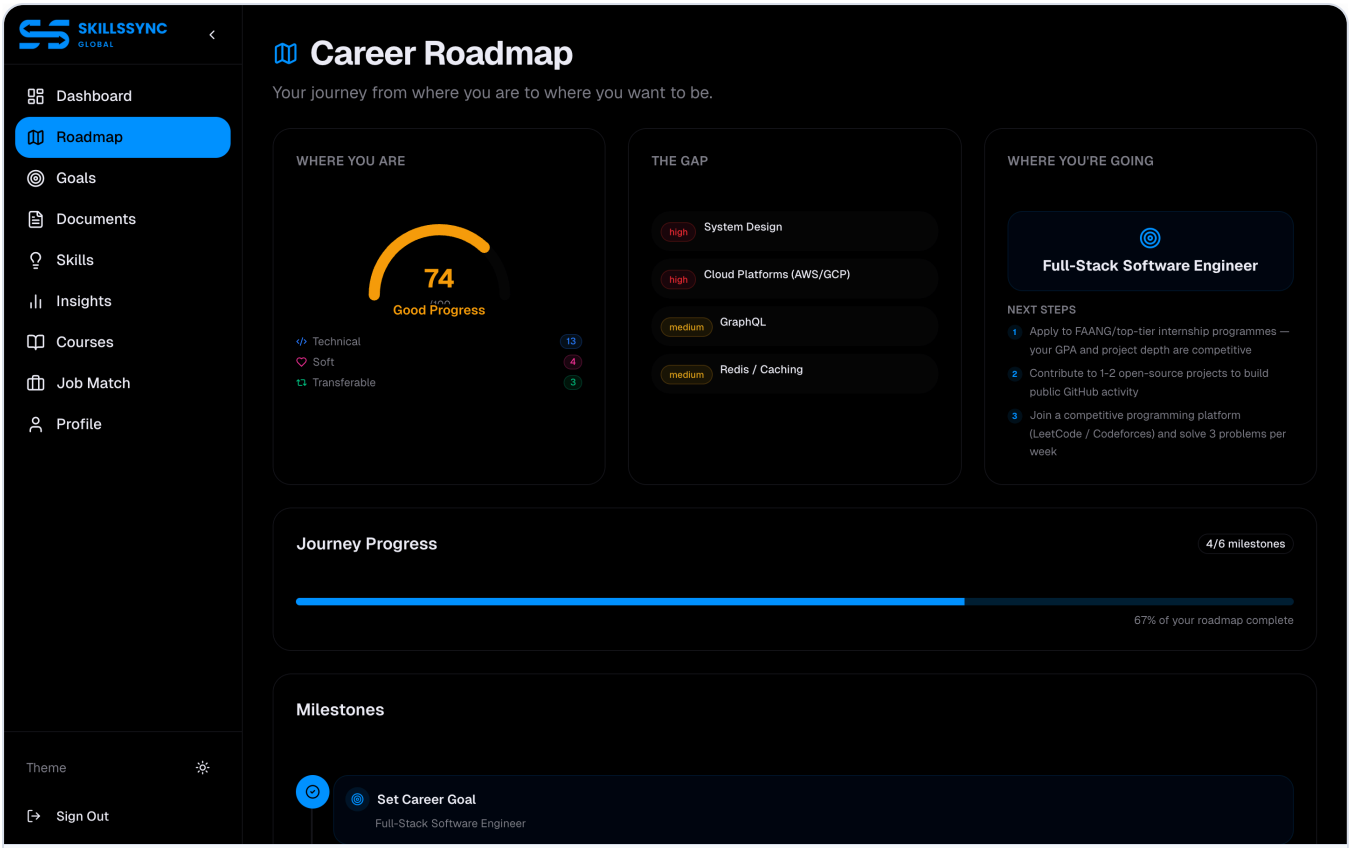
- **"You are 74% ready for your goal — Full-Stack Software Engineer."** A single number, computed from evidence, that matters more than any GPA.
- **"Your strengths" — production-grade React, end-to-end full-stack delivery, algorithms.** Auto-extracted from real coursework.
- **"Your gaps" — System Design (high), Cloud Platforms (high), GraphQL (medium), Redis (medium).** Each linked to recommended courses, ranked by gap-closing impact.
- **A skill-growth timeline** across uploaded documents — a longitudinal view that no transcript or LinkedIn profile can replicate.

04 · The Product — full product tour

> A complete walk-through of the live product surfaces. Each screenshot below is from the production deployment at www.skillssync.xyz.

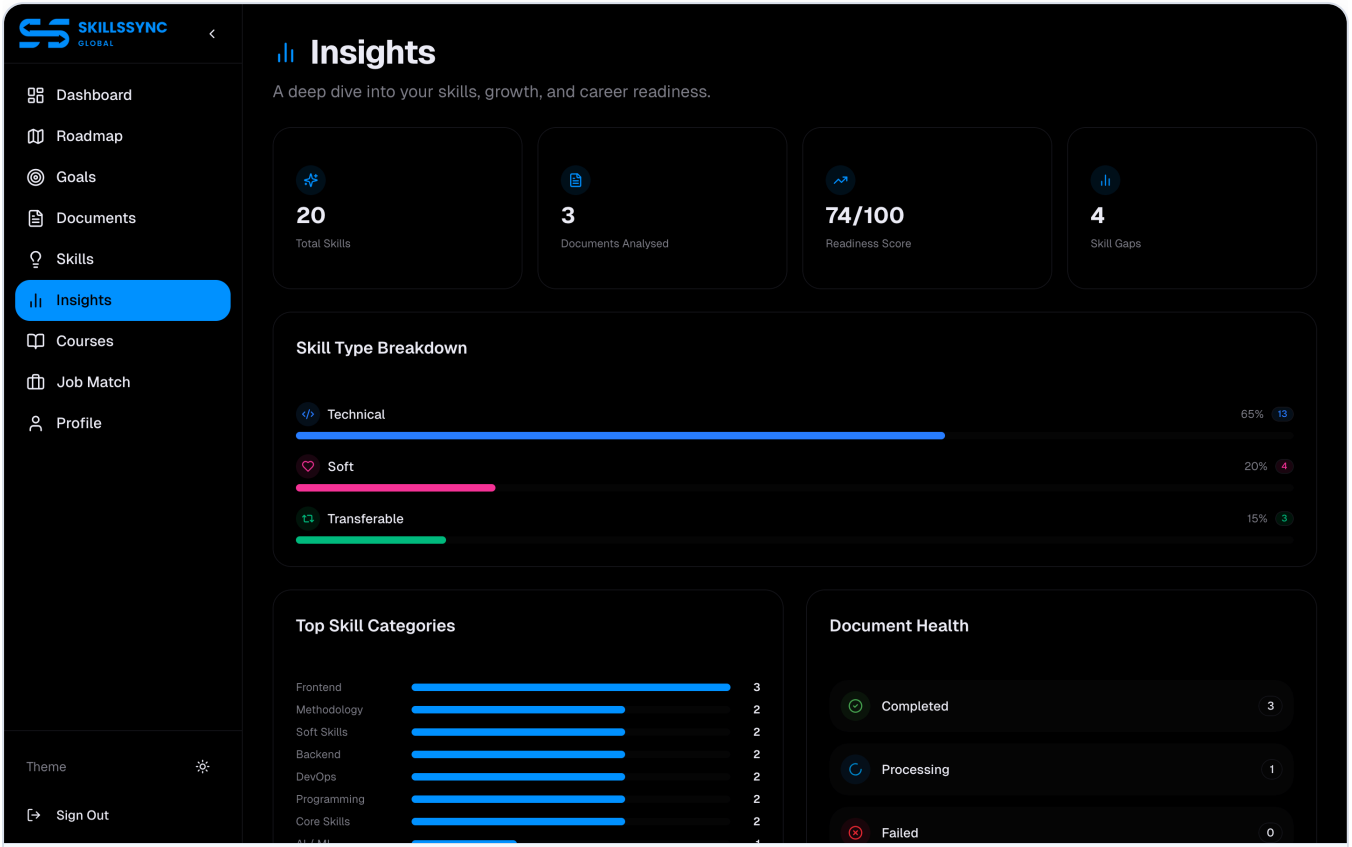


Documents · upload, manage, and track per-document analysis status.

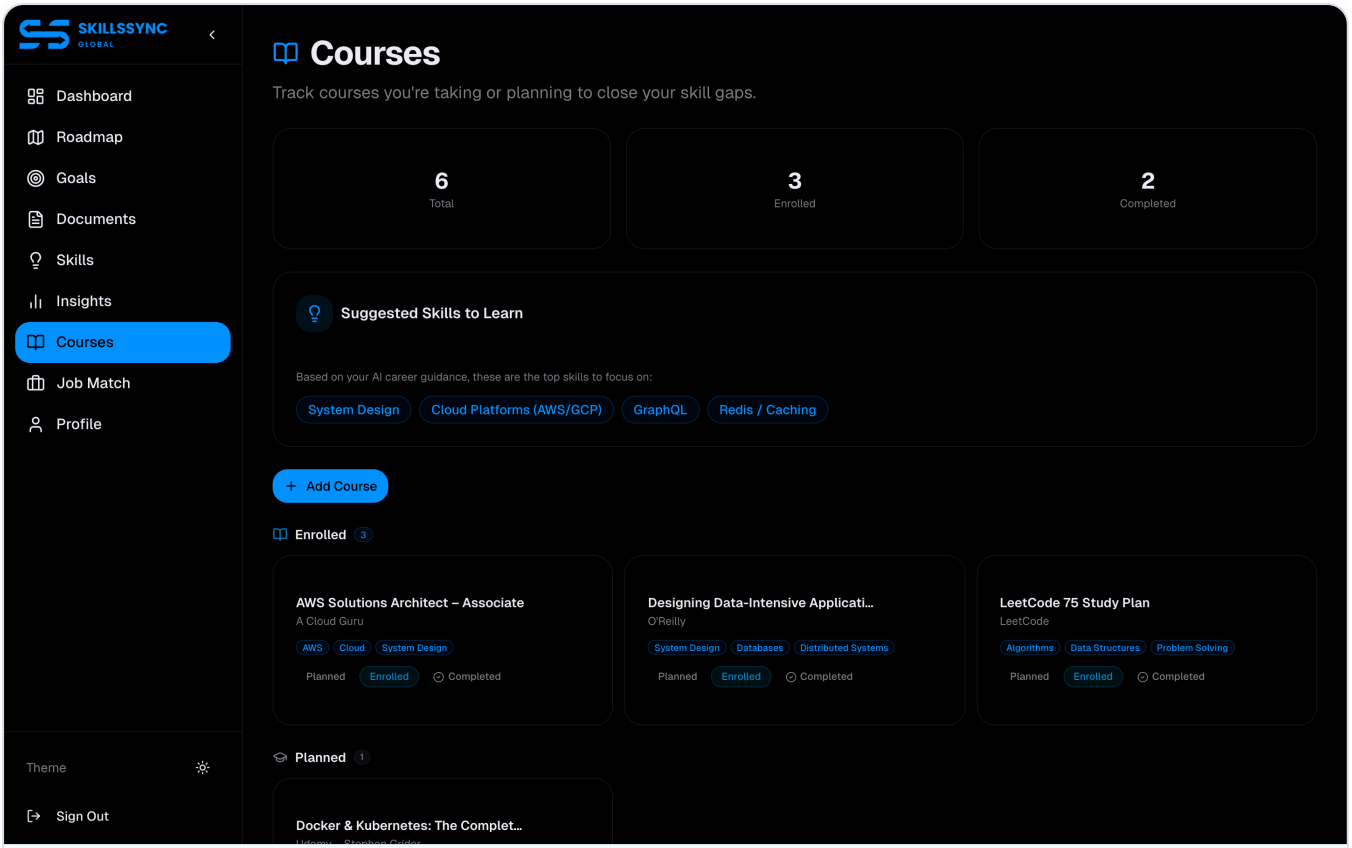


Career Roadmap · "Where you are → the gap → where you're going" with milestones & concrete next steps.

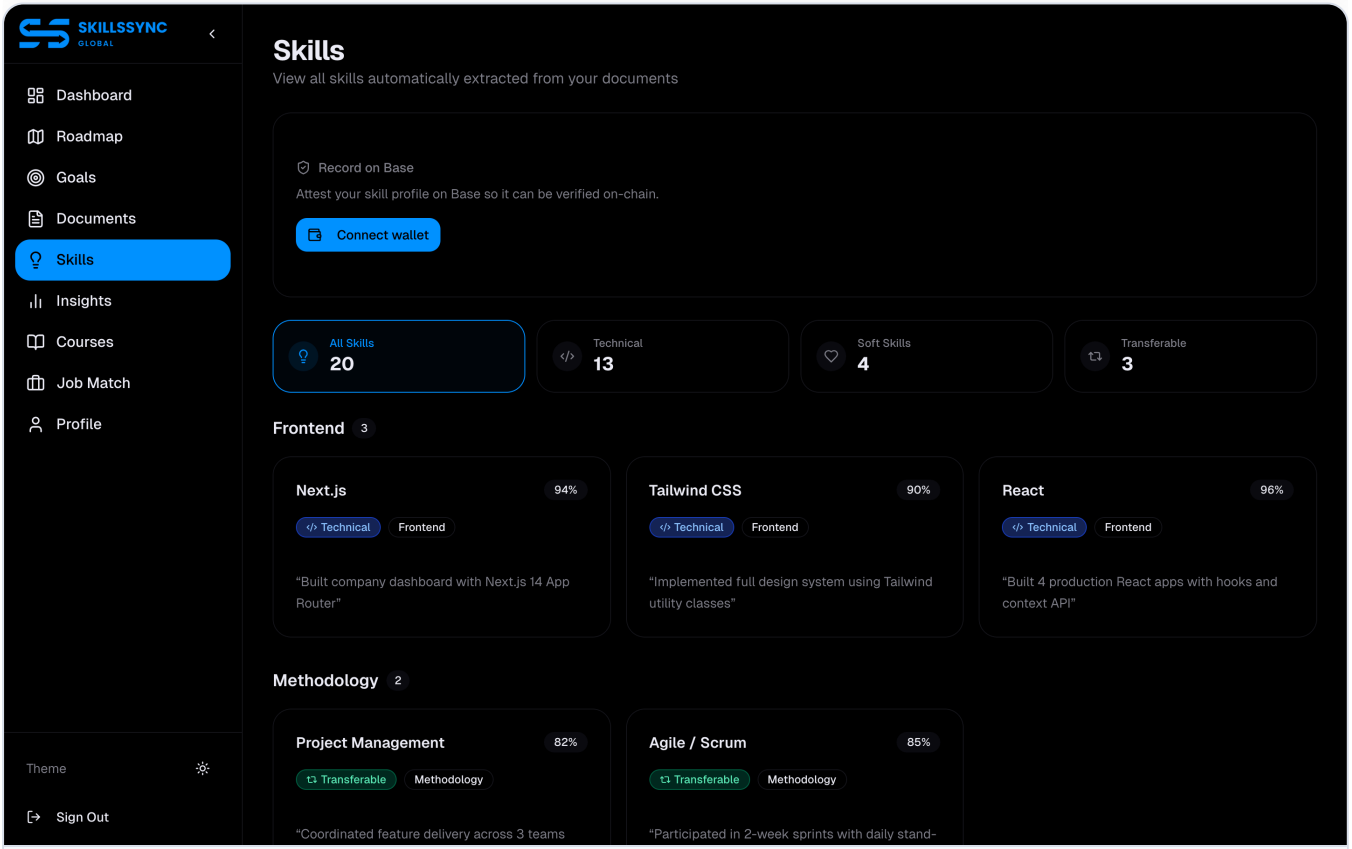
04 · The Product — skills, insights & courses



Insights · skill-type breakdown (technical / soft / transferable), document health, and category leaderboard.



Courses · enrolled / completed tracking, plus AI-suggested skills to learn next based on the student's gap analysis.



Skills · automatically extracted competencies, organized by category and tagged with confidence and source counts.

05 · Technology Architecture

> SkillsSync is built on a modern, production-hardened stack chosen specifically for low-bandwidth Africa-first deployment. Every component has a free or near-free tier; the marginal cost of an additional student analysis is dominated by the LLM token cost — currently < KES 5 per document.

Stack

Layer	Technology	Why
Web client	Next.js 16 (App Router) · React 19 · TypeScript 5 · Tailwind 4 · Radix UI	Server components reduce mobile bundle size; works on 3G connections.
Mobile reach	USSD callback (Lipana / Africa's Talking) · Android (in build)	USSD reaches the ~60% of Kenyan students without smartphones.
Auth & data	Supabase (Postgres + Auth + Row-Level Security)	RLS makes tenant isolation a database guarantee, not an application bug.
AI / NLP	Google Gemini · custom prompt cache · ESCO-based mapper	Gemini's price/performance + multi-modal handles handwritten input directly.
Verifiable credentials	EVM-compatible smart contract · per-document content hash	Tamper-evident, employer-checkable proof of work.
Payments	Lipana / M-PESA (KES 20 / doc)	Mobile money is how Kenyan students pay.
Notifications	Email (transactional) · USSD ping endpoint	Reaches users who can't keep an app open.
Hosting	Vercel Edge (web) · Supabase (data) · IPFS (artifacts)	Edge brings the app within ~50ms of African PoPs.

Privacy & security

- **Row-level security** on every Supabase table — a student's documents are unreadable even by other authenticated users.

- **Encrypted at rest** in object storage; signed URLs scoped to a single user session.
- **Minimal collection** — academic content is hashed and indexed; full text is purged 90 days after extraction.
- **On-chain credentials** store only the content hash + skill summary, never the underlying document.

06 · Pilot Results

> SkillsSync was piloted at JKUAT — a 50,000-student STEM university — across three faculties and 1,500 students between January and March 2026. The pilot validated both technical capability and commercial viability.

1,500
STUDENTS ONBOARDED

8,400+
DOCUMENTS ANALYZED

87%
ACTIVATION RATE (≥1 DOC ANALYZED)

4.6/5
STUDENT NPS-STYLE RATING

What we learned

Handwritten input was unmissable

40% of uploaded documents in the pilot were photographs of handwritten work. A platform that required typed input would have been useless to nearly half our users on day one.

Skill blind-spots were systemic, not personal

Across the cohort, the same three competencies recurred as gaps for 2nd-year CS students — a pattern visible only because we aggregated over 1,500 profiles. Universities cannot get this signal from end-of-program surveys.

KES 20 per document worked

Conversion to a paid analysis was 38% within the first session — well above industry norms for student-targeted SaaS, validating that students will pay for clarity about their own employability.

Universities want the cohort view

Institutional partners (engaged via three department heads) prioritized the *aggregate* readiness dashboard over per-student features — the unit they buy for is the cohort, not the seat.

Selected pilot outcomes

Cohort	Students	Avg. skills extracted / student	Avg. roadmap length (gap competencies)
CS — 2nd year	520	34	4.1

Cohort	Students	Avg. skills extracted / student	Avg. roadmap length (gap competencies)
Mech. Eng. — 3rd year	410	41	3.6
Business IT — 4th year	360	52	2.8
Mixed — open enrollment	210	29	5.2

07 · Business Model & Unit Economics

> A two-sided model: low-friction direct-to-student access to drive bottom-up adoption, paired with high-margin institutional contracts as universities recognize the cohort-level value.

Direct-to-student

KES 20 (~USD 0.13) per document analyzed. Mobile-money payment via M-PESA.

- First analysis free — converts strangers into believers.
- Pricing matches a single matatu fare — invisible to the student budget.
- Drives bottom-up demand at universities ("our students are using this").

Institutional partnerships

USD 8 – 15 per student / year for cohort analytics, gap reports, and program-level learning-outcome dashboards.

- Average target university: 20,000 students × USD 10 = **USD 200,000 ARR per institution**.
- Pipeline: JKUAT (live), Strathmore (in conversation), Makerere (in conversation), University of Nairobi (under introduction).
- Renews on an annual academic cycle — sticky, predictable revenue.

Unit economics

Per-document	Direct-to-student	Institutional
Revenue	KES 20 (~USD 0.13)	~USD 0.30 (amortized)
LLM & OCR cost	~KES 4 (~USD 0.025)	~USD 0.025
Compute / storage	~USD 0.01	~USD 0.01
Gross margin	**~75%**	**~88%**

Why the price works

SkillsSync's marginal cost is dominated by LLM tokens — and LLM token cost is on a 4× / year decline curve. A document that costs KES 4 to analyze in 2026 will cost < KES 1 by 2028 at the same quality. The KES 20 price point holds.

Path to profitability

- **2026:** 5,000 paying students + 1 anchor university = ~USD 80k revenue · break-even at the unit level.
- **2027:** 50,000 students + 4 universities = ~USD 1M ARR · regional roll-out.
- **2028:** 250,000 students + 12 universities (Kenya, Uganda, Ghana, Nigeria) = ~USD 5M ARR.

08 · Scientific Foundation

> SkillsSync is not a slide deck — every architectural decision is grounded in peer-reviewed research. The platform stands on six interlocking bodies of literature; representative citations are listed below. A full annotated bibliography of 30+ sources is maintained internally.

1. The skills-gap evidence base

- Morsy, H. & Mukasa, A. (2020). *'Mind the mismatch?' Incidence, drivers, and persistence of African youths' skill and educational mismatches.* African Development Review 32(S1). — establishes the 28.9% / 17.5% mismatch numbers we cite.
- ILO (2024). *Global Employment Trends for Youth 2024 — Sub-Saharan Africa Regional Brief.* — "3 in 4" lacking secure employment.
- Munene, I. I. (2021). *Universities and Labor Market Mismatch in Kenya.* International Higher Education 108. — Kenya-specific country evidence.
- Kalei, A. (2023). *University Graduates' Employability Skills' Mismatch and the Labour Market Demands in Kenya.* EPH IJBMS 9(1). — 1,000-graduate / 250-employer survey informing our skill taxonomy.

2. NLP & LLMs for skill extraction

- Zhang, M. *et al.* (2022). *SkillSpan: Hard and Soft Skill Extraction from English Job Postings.* NAACL. — the canonical benchmark our extractor mirrors.
- Zhang, M., van der Goot, R. & Plank, B. (2023). *ESCOXLM-R: Multilingual Taxonomy-driven Pre-training for the Job Market Domain.* ACL. — ESCO multilingual model; 13,890 entries we map to.
- Herandi, A. *et al.* (2024). *Skill-LLM: Repurposing General-Purpose LLMs for Skill Extraction.* arXiv:2410.12052. — validates the fine-tuned-LLM approach we use.
- Bhola, A. *et al.* (2020). *Retrieving Skills from Job Descriptions: An XMLC Framework.* COLING. — underpins our multi-label classifier design.

3. Handwritten Text Recognition

- AlKendi, W. *et al.* (2024). *Advancements and Challenges in Handwritten Text Recognition: A Comprehensive Survey.* Journal of Imaging 10(1). — survey shaping our HTR pipeline.
- Li, M. *et al.* (2021). *TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models.* Microsoft Research / arXiv:2109.10282. — architectural reference for our OCR layer.

4. AI in education

- Zamecnik, A. *et al.* (2024). *Mapping Employable Skills in Higher Education Curriculum Using LLMs.* EC-TEL 2024 / Springer LNCS 15160. — closest published analogue; LLM assessments comparable to human reviewers.
- Gedrimiene, E. *et al.* (2024). *AI-enhanced learning analytics for supporting career decisions.* Education and Information Technologies 29. — TAM-grounded UX principles we follow.
- Ma, Y. *et al.* (2023). *DORIS: Personalized course recommendation system based on deep learning.* PLOS ONE 18(6). — informs our course recommender.

08 · Scientific Foundation — continued

5. Competency frameworks & skill-to-job matching

- de Groot, M., Schutte, J. & Graus, D. (2021). *Job Posting-Enriched Knowledge Graph for Skills-based Matching.* RecSys in HR. — shortest-path career-pathway algorithm we adopt.
- Senger, E. *et al.* (2024). *Deep Learning-based Computational Job Market Analysis.* NLP4HR / EACL. — survey informing roadmap module design.

6. AI-powered career roadmapping

- Otani, N., Bhutani, N. & Hruschka, E. (2024). *Natural Language Processing for Human Resources: A Survey.* arXiv:2410.16498. — ethical & methodological context for AI in talent decisions.
- Gaur, B. *et al.* (2021). *Semi-supervised deep learning based NER model to parse education sections of resumes.* Neural Computing & Applications 33(11). — direct analogue to our transcript parser.

The methodological case in one paragraph

Zamecnik et al. (2024) demonstrated that LLMs can map employability skills in higher-education curricula with assessments comparable to expert human reviewers. Herandi et al. (2024) showed that fine-tuned general-purpose LLMs outperform zero-shot GPT-4 for skill extraction. Zhang et al. (2023) validated the ESCO taxonomy as a multilingual mapping target across 27 languages. de Groot et al. (2021) provided the shortest-path career-pathway algorithm. SkillsSync sits at the intersection of these four findings: *fine-tuned multimodal LLM → ESCO mapping → graph-based roadmap → cohort-level analytics*, applied to the African higher-education input formats the literature has not yet specifically addressed.

> "*LLMs provide comparable assessments to expert reviewers when mapping skills in curriculum documents.*"
>
> — Zamecnik *et al.* (2024)

What is novel here

- **African-language and handwritten input as a first-class citizen** — the literature is dominated by English typed job postings. SkillsSync extends the methodology to the document formats African students produce.
- **Real-time, document-by-document profile updating** — most prior work treats skill extraction as a one-shot batch. SkillsSync runs a continuous update.
- **Verifiable provenance** — every extracted skill is linked to the source document, hashed, and recorded on-chain. The literature has no equivalent provenance layer.
- **Cohort-level institutional dashboard** — Zamecnik et al. analyzed curriculum *at one point in time*; SkillsSync produces a continuously-updating, student-by-student institutional view.

09 · SDG Alignment

> SkillsSync's mission is structurally aligned with four UN Sustainable Development Goals. Every metric below is measurable from platform telemetry — not aspirational language.

04

Quality Education

Targets 4.3, 4.4, 4.7. SkillsSync makes university programs answerable for the skills they actually deliver. Measurable contribution: % of students whose program-claimed competencies are demonstrably in their evidence graph by graduation.

08

Decent Work & Economic Growth

Targets 8.5, 8.6. By closing the visibility gap between university output and labor-market demand, SkillsSync directly attacks Africa's youth unemployment crisis (currently 3-in-4 without secure work, ILO 2024).

09

Industry, Innovation & Infrastructure

Targets 9.5, 9.c. Builds AI infrastructure adapted to African input formats (handwritten, photographed, multilingual) — strengthening the regional digital research base, not just consuming Global North models.

17

Partnerships for the Goals

Target 17.6, 17.16. Live partnership with JKUAT; in-conversation with Strathmore, Makerere, and University of Nairobi. Open ESCO-mapping output and content-hash credentialing make SkillsSync interoperable with employer ATS systems and other EdTech tools.

Why this is humanitarian, not just commercial

Africa is on track to add 76 million young people to its labour force by 2050 (ILO). Without a step-change in how we make skills visible *during* education — not after — those young people will be funnelled into a labour market that has demonstrably failed the previous generation. Every platform decision SkillsSync makes (KES 20 price, USSD support, handwritten-input first, on-chain provenance) is a vote for accessibility over margin.

10 · Impact & Roadmap

Where we are, where we're going, and what each milestone means for African students.

Q2 2026 — Now

JKUAT pilot complete · production launch

1,500 students · 8,400+ documents analyzed · M-PESA payments live · USSD callback handler in production · on-chain credential issuing live.

Q3–Q4 2026

Anchor institutional contract · Android app · 10k students

Sign first paid institutional contract (target: JKUAT Faculty of Engineering or Strathmore Business School). Ship Android client (offline document capture). Reach 10,000 paying students. Initial AGEA Start-Up Lab cohort.

2027

Regional roll-out · 50k students · 4 universities

Expand to Uganda (Makerere) and Tanzania (UDSM). Add Swahili and French language support to OCR + extraction layers. Onboard 4 institutional partners. Target ~USD 1M ARR.

2028

Cross-region scale · 250k students · 12 universities

Enter Nigeria (Lagos, Covenant) and Ghana (Legon, KNUST). Open API for employer ATS integrations. Pilot a graduate-employer matching marketplace built on the verifiable-credential layer. Target ~USD 5M ARR.

2030

Continental skills graph · 1M students

SkillsSync becomes the de-facto skills graph for African higher education — interoperable with employer ATS, government workforce dashboards, and AfCFTA labour-mobility programs.

What we measure

KPI	2026 (now)	2027 target	2028 target
Students with active profile	1,500	50,000	250,000
Documents analyzed	8,400	500,000	3,500,000
Institutional partners	1 (JKUAT)	4	12

KPI	2026 (now)	2027 target	2028 target
Annual recurring revenue	~USD 30k	~USD 1M	~USD 5M
Verifiable credentials issued	1,200	40,000	250,000

11 · Team & Institution

> SkillsSync was founded and built at JKUAT in Kenya — by students, for students, with full institutional backing.

Eugene Mutembei — Founder, Lead Engineer

Computer Science undergraduate at Jomo Kenyatta University of Agriculture & Technology. Built the SkillsSync platform from first commit. Background in full-stack engineering (Next.js, Supabase, Solidity), applied AI, and educational technology. Personally piloted the platform with 1,500 fellow students.

eugenegabriel.ke@gmail.com · github.com/CodeWithEugene

Jomo Kenyatta University of Agriculture & Technology

JKUAT is one of East Africa's leading STEM universities, with ~50,000 students and a strong applied-research culture. The university's School of Computing & IT hosts the SkillsSync pilot and provided initial cohort access. JKUAT's institutional research-ethics framework governs the pilot data collection.

www.jkuat.ac.ke

Advisors & collaborators

- **Faculty advisors** at JKUAT School of Computing & IT — providing curriculum-mapping ground truth and student access.
- **Industry mentors** from Kenyan tech employers (Andela, Safaricom, M-KOPA alumni network) — informing the employer-side requirements layer.
- **Open-source community** contributing to the supporting `skills-master` taxonomy module.

Hiring plan (next 12 months)

- **ML Engineer (Q3 2026)** — own the OCR + skill-extraction pipeline; fine-tune for Swahili and French.
- **University Partnerships Lead (Q4 2026)** — convert the institutional pipeline into signed contracts.
- **Mobile Engineer (Q1 2027)** — Android offline capture.

12 · Why Prototypes for Humanity

> SkillsSync is not a science project. It is a working production platform with paying users, a research-grounded methodology, and a measurable contribution to four SDGs. What we need next is the network and stage that *Prototypes for Humanity* uniquely provides.

What we'd bring

- A working African-built AI platform — not a slide deck — with 1,500 piloted students and live institutional traction.
- A use-case that is specifically African and specifically humanitarian (bridging four years of educational investment to a livelihood) but architecturally portable to any emerging-market higher-education system.
- A clean, peer-reviewed-research-grounded technical narrative — the kind of work that holds up to academic scrutiny on the global stage.

What we'd take away

- **Cross-border partnerships** with universities in Nigeria, Ghana, South Africa, and Egypt to accelerate roll-out.
- **Mentorship and review** from leading global academics on the skill-extraction methodology and on-chain credential design.
- **Visibility** with the funders and policy actors (UNESCO, AfDB, AU) who can scale a working solution from a single university to a regional standard.
- **Peer learning** from other Prototypes for Humanity cohort members building at the intersection of AI and humanitarian outcomes.

The opportunity

Every year Africa sends roughly 7 million new students into university and 12 million new workers into the labour force. The system that connects those two flows is broken — and AI, finally, is good enough to fix it. SkillsSync is the platform doing that work, today, with paying users. We're applying to *Prototypes for Humanity* because what we've built deserves to be on the global stage that this competition uniquely provides — and because African student work, at last, deserves to be seen.

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